Beta-reduction is exactly the transitive closure of parallel beta-reduction

The Open Logic Project contributors

openlogic:lambda-calculus/church-rosser:local-snapshot-20261013;
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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: both inclusions are written out with their chain displays and their two named citations inside the span; the only quoted-notation line with a small author typo ('Since is xred contains') is delivered as is.

Editorial note. The label lem:str names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Editorial note. OLP reuses this label inside the unit; the card is identified by its statement span (source lines 524-526) and the claim key carries the statement start line.

Allowed standard prerequisites (editorial).

- Single beta steps are parallel steps. Same unit, \ollabel{lem:one-par} (source line 482). First inclusion: each bredone link of a reduction sequence is replaced by a bredpar link.
- Parallel steps are reduction sequences. Same unit, \ollabel{lem:par-red} (source line 490). Second inclusion: each bredpar link is expanded into a finite sequence of single steps.

Source reference. Original lemma lem-one-par: sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex, source lines 486–486. Prerequisite pointer invoked at proof line(s) 533; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma lem-par-red: sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex, source lines 496–496. Prerequisite pointer invoked at proof line(s) 538; it names a standard fact of the same unit that the

counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *$\to_\beta$ is the smallest transitive relation containing $\to_\beta$.*

Proof. Let $\to$ be the smallest transitive relation containing $\to_\beta$.

$\to_\beta \subseteq \to$: Suppose $M \to_\beta M'$, i.e., $M\,\mathrm{id}\,M_1 =_\beta \cdots =_\beta M_k\,\mathrm{id}\,M'$. By lem-one-par, $M\,\mathrm{id}\,M_1 \to_\beta \cdots \to_\beta M_k\,\mathrm{id}\,M'$. Since is $\to$ contains $\to_\beta$ and is transitive, $M \to M'$.

$\to \subseteq \to_\beta$: Suppose $M \to M'$, i.e., $M\,\mathrm{id}\,M_1 \to_\beta \cdots \to_\beta M_k\,\mathrm{id}\,M'$. By lem-par-red, $M\,\mathrm{id}\,M_1 \to_\beta \ldots \to_\beta M_k\,\mathrm{id}\,M'$. Since $\to_\beta$ is transitive, $M \to_\beta M'$. □